

is optimised to achieve extended battery life. The device features a powerful 16-bit RISC CPU, 16-bit registers, and constant generators that contribute to the maximum code efficiency. The digitally controlled oscillator (DCO) allows wake-up from low-power modes to active mode in less than 1 μ s.

This ultra-low-power MCU is equipped with two built-in 16-bit timers, two universal serial communication interfaces (USCs), 10-bit ADC with integrated reference and data transfer controller, two general-purpose operational amplifiers and 32 input/output (I/O) pins.

The MCU has 32 kB of Flash memory and 1 kB of RAM on chip. It is a battery-driven (coin-cell-driven) system. Therefore the long battery life (two-three years) is the main design challenge. Also, the system should be compact enough to be worn as a ring in the finger or as a bracelet in the wrist. The MCU supports ultra-low power modes and the transmitter has a very small PCB footprint, which makes it suitable for this 3-D gestures and character recognition application.

3-D gesture and character recognition system

The implementation of 3-D gesture and character recognition consists of four distributed components: 3-D gesture transmitter module, 3-D gesture receiver module, 3-D gesture and characters recognition module and host application.

3-D gesture transmitter module.

Fig. 1 shows the block diagram of the 3-D gesture transmitter module. It consists of an MEMS motion sensor (LIS302DL) and an RF transceiver (CC2500) interfaced with an MSP430F2274 MCU using I²C and SPI interfaces, respectively.

A universal serial communication interface (USCIB) of the MSP430F2274 is configured as SPI to communicate with the RF transceiver (CC2500) and GDO2 pin of CC2500 generates an interrupt at port pin P2.7 of



Fig. 1: Block diagram of 3-D gesture transmitter module

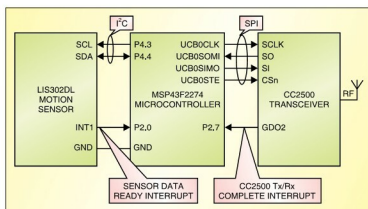


Fig. 2: Hardware configuration of the 3-D gesture transmitter module

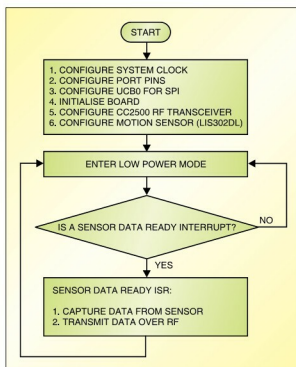


Fig. 3: Flow-chart for the 3-D gesture transmitter module



Fig. 4: Block diagram of the 3-D gesture receiver module

MSP430F2274 to indicate that the initiated transmission or reception activity has been completed.

Port pins P4.3 and P4.4 are used for I²C protocol emulation and interfaced with I²C lines (SDA and SCL) of the motion sensor (LIS302DL). INT1 line of

the motion sensor (LIS302DL) interrupts the MSP430F2274 at port pin P2.0 to indicate that a new set of 3-axis acceleration data is available. The detailed hardware configuration is shown in Fig. 2.

The basic firmware flow is shown in Fig. 3. Initially, the motion sensor, RF transceiver and

board-specific parameters are configured. The MCU enters a very low-power mode (LPM3) and waits in a super loop for a sensor-data-ready interrupt. Whenever a sensor-data-ready interrupt occurs, it wakes up, captures data from the motion sensor, transmits it wirelessly using the RF transceiver and returns to low-power mode.

3-D gesture receiver module. The 3-D gesture receiver module is built around an MSP430F2274 MCU, interfaced with the RF transceiver (CC2500) using SPI and with the host system (Linux based) using a universal serial bus (USB 2.0) interface. The block diagram of the module is shown in Fig. 4.

Similar to the transmitter module, a universal